

Dharani N. Kesh. r.v
192424201

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2, BL6)

Assessment Tool Weightage: 30%

83.1/100

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected

1. o the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes. t

ANSWER:

A **star topology** is an ideal choice for a 20-person startup because every device is directly connected to a central network switch. A 24-port managed Layer 2/Layer 3 switch can be installed at the center of the network. The extra ports allow the company to add more computers, printers, IP phones, or other devices in the future.

Each workstation can be connected to the switch using **Cat6 UTP cables**, which provide reliable Gigabit Ethernet connectivity for typical office distances. The central switch can be connected to a router or firewall that provides internet connectivity, DHCP services, NAT, and network security.

A wireless access point can also be connected to the switch to provide Wi-Fi connectivity for laptops, smartphones, and tablets. VLANs can be configured to separate employee computers, guest Wi-Fi, VoIP phones, and other devices.

The major advantage of star topology is **fault isolation**. If the cable connecting one employee's computer fails, only that computer loses connectivity while the remaining employees continue working normally.

Advantages:

- Easy installation and centralized management
- Simple troubleshooting and maintenance
- Failure of one cable affects only one device
- Easy to expand the network
- Supports high-speed communication
- Allows VLAN-based network segmentation

Conclusion:

Star topology provides a reliable, secure, manageable, and scalable network solution for a small startup.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

ANSWER:

A banking organization requires extremely high network availability because even a short network interruption can affect online banking, ATMs, financial transactions, and internal banking applications. Therefore, a **partial mesh topology** is suitable for connecting important core network devices.

In this design, core routers, switches, firewalls, and important servers have multiple connections. If one network link becomes unavailable, traffic can use another available path. This provides redundancy and prevents a single link failure from bringing down the complete network.

Enterprise-grade routers can provide redundant internet connections. Routing protocols such as **OSPF and BGP** can be used to select alternative routes automatically when a failure occurs.

Fiber-optic cables are preferred for core connections because they provide high bandwidth, low latency, long-distance communication, and immunity to electromagnetic interference.

Additional reliability can be provided using redundant power supplies, UPS systems, backup servers, and RAID storage.

Advantages:

- Very high reliability
- Multiple communication paths
- Automatic rerouting during failures
- Reduced downtime
- High-speed fiber connectivity
- Suitable for critical financial applications

Conclusion:

A partial-mesh network with redundant devices and communication links provides the fault tolerance and availability required for modern banking infrastructure.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8-

or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

ANSWER:

For a small classroom laboratory with a limited budget, a **bus topology** can be considered because multiple computers share a common communication medium. It requires less infrastructure compared with more complex redundant topologies.

Traditional bus networks use a coaxial backbone cable with connectors and terminators at both ends. However, this technology is now largely obsolete for modern Ethernet networks.

For a practical modern classroom, an inexpensive unmanaged Ethernet switch with Cat5e or Cat6 cables would be a better implementation. It provides the simplicity and low cost expected from a small lab while avoiding many limitations of traditional bus networks.

In a traditional bus topology, if the main backbone cable develops a fault, communication between several or all computers may be interrupted. Shared-medium communication can also result in collisions when multiple computers transmit simultaneously.

Advantages:

- Low installation cost
- Simple network structure
- Requires fewer networking devices
- Suitable for small networks with light traffic

Limitations:

- Backbone failure can affect many computers
- Traditional bus Ethernet can experience collisions
- Troubleshooting can be difficult
- Network performance decreases as traffic increases

- Not suitable for large-scale expansion

Conclusion:

Bus topology can be considered for a low-cost theoretical classroom network, but a small switched-star network is the recommended practical solution for modern Ethernet environments.

4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need

for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

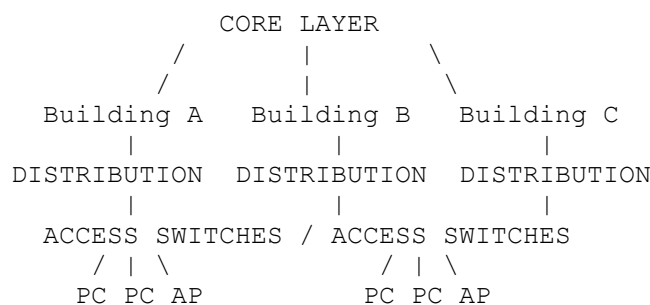
ANSWER:

large university contains thousands of users distributed across classrooms, laboratories, libraries, hostels, administrative buildings, and research centers. Therefore, using only one topology would not be efficient. A **hybrid topology** combines different topologies to meet different requirements.

The campus backbone can use a **fiber-based ring or partial-mesh structure** between core routers and multilayer switches. Redundant fiber links ensure that a single cable or device failure does not disconnect an entire building.

Each building can have a distribution switch connected to the campus core using high-speed fiber uplinks. Within each building, a star topology can be used to connect floor-level access switches and end devices.

The network can follow the three-layer architecture:



Wireless access points can be connected through **Power over Ethernet (PoE)**, allowing both power and data to travel through the same Ethernet cable.

New buildings can be added by connecting their distribution switches to the existing campus backbone without redesigning the entire network.

Advantages:

- Highly scalable
- Supports thousands of users
- Provides backbone redundancy
- Easy to manage building by building
- Supports both wired and wireless devices
- Easy future expansion
- High-speed fiber communication

Conclusion:

A hybrid hierarchical topology provides the university with a combination of reliability, scalability, performance, and easy network management.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

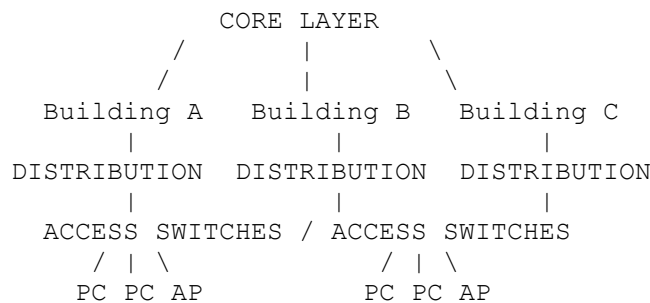
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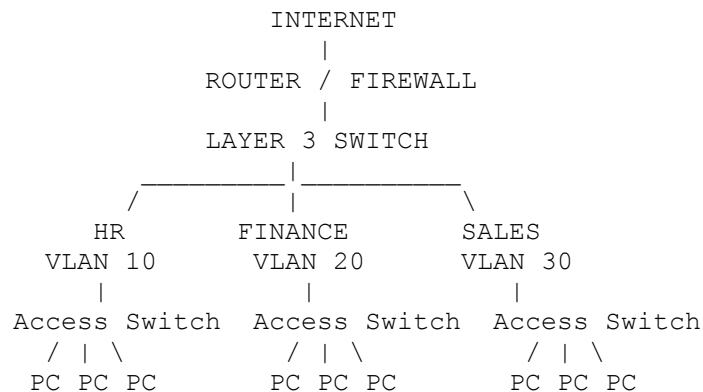
5. Network Design for a Company with Multiple Departments

Recommended Design: Hierarchical Network with VLANs

A company containing departments such as **HR, Finance, Sales, and IT** requires a structured network to ensure efficient communication and security. A hierarchical architecture with access switches, a Layer 3 switch, routers, and VLANs is suitable.

Each department can be assigned a separate VLAN. For example, HR can use VLAN 10, Finance VLAN 20, Sales VLAN 30, and IT VLAN 40. This logically separates departmental traffic even when the departments use the same physical network infrastructure.

A typical structure can be represented as:



The Layer 3 switch performs **inter-VLAN routing**, allowing communication between departments when permitted by security policies. ACLs can be used to restrict sensitive resources. For example, Finance systems can be prevented from being accessed by unauthorized users.

Subnetting can also be used to assign different IP address ranges to each department. Redundant links and backup network devices can be introduced for improved reliability.

Advantages:

- Better network security
- Reduced broadcast traffic
- Easy departmental management
- Efficient IP address allocation
- Supports inter-department communication
- Easy to expand as the company grows
- Allows centralized monitoring and control

Conclusion:

A hierarchical network using VLANs, Layer 3 switching, routing, and firewall protection provides a secure, efficient, scalable, and manageable solution for companies with multiple departments.

RUBRICS FOR CALCULATION

	Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
B	Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
C	Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
D	Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
E	Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
F	Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
G	Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
H	Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

	A	B	C	D	E	F	G	H	I	total	
Q ₁	2	3	2	2	1	2	1	1	0.5	0.5	15
Q ₂	3	4	2	2	2	2	1	0.5	0.5	1	18
Q ₃	2	4	3	1	2	2	0.5	0.5	1	0.5	16.5
Q ₄	3	3	4	1	1	1	0.5	1	1	1	16.5
Q ₅	2	4	3	1	2	1	1	1	1	1	17

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